

BookIQ API

Overview

A data-driven RESTful book metadata intelligence platform. Provides full CRUD, PostgreSQL full-text search with GIN indexing, hybrid Reciprocal Rank Fusion search, content-based and collaborative recommendations, genre trend analytics, and an MCP server layer for AI agent integration.

Authentication: JWT Bearer tokens. Obtain a token via POST `/api/auth/login/`. Include as: Authorization: Bearer <token>

Roles: reader → curator → admin (ascending permissions).

Tags

auth

Registration, login, JWT token management

books

Book CRUD operations

search

Keyword, full-text, and hybrid search

recommendations

Personalised book recommendations

ratings

Book rating and review management

imports

CSV bulk import pipeline

analytics

Genre trends and catalogue health

Paths

***POST* /api/auth/register/ Register a new user**

Creates a new user account. Default role is reader.

Responses

Code	Description	Links
201	User created with JWT tokens	No Links
400	Validation error (duplicate email/username, password mismatch)	No Links

Type	Name	Scopes
apiKey	jwtAuth	

POST /api/auth/login/ Login and obtain JWT tokens

Authenticates credentials and returns access + refresh tokens. Access token expires in 60 minutes. Refresh token expires in 7 days.

Responses

Code	Description	Links
200	Access and refresh tokens with role claim	No Links
401	Invalid credentials	No Links

POST /api/auth/token/refresh/

Takes a refresh type JSON web token and returns an access type JSON web token if the refresh token is valid.

Responses

Code	Description	Links
200	Content application/json	No Links

GET /api/auth/profile/ Get or update user profile

Responses

Code	Description	Links
200	User profile data	No Links

Type	Name	Scopes
apiKey	jwtAuth	

PATCH /api/auth/profile/ Get or update user profile

Responses

Code	Description	Links
200	User profile data	No Links

Type	Name	Scopes
apiKey	jwtAuth	

GET /api/books/ List all books

Returns a paginated list of all books ordered by creation date. Supports filtering via BookFilter params.

Parameters

Type	Name	Description	Schema
query	published_before <i>optional</i>	Published year to	integer
query	author <i>optional</i>	Author (partial match)	string
query	ordering <i>optional</i>	Which field to use when ordering the results.	string
query	language <i>optional</i>	Language code (e.g. en)	string
query	max_quality <i>optional</i>	Maximum quality score (0.0 - 1.0)	number (float)
query	q <i>optional</i>	Keyword search	string
query	search <i>optional</i>	A search term.	string
query	min_quality <i>optional</i>	Minimum quality score (0.0 - 1.0)	number (float)
query	min_rating <i>optional</i>	Minimum average rating (1 - 5)	number (float)
query	genre <i>optional</i>	Genre (partial match)	string
query	publisher <i>optional</i>	Publisher (partial match)	string
query	page <i>optional</i>	A page number within the paginated result set.	integer
query	published_after <i>optional</i>	Published year from	integer

Type	Name	Description	Schema
query	is_flagged <i>optional</i>	Is flagged for review	boolean
query	page_size <i>optional</i>	Number of results to return per page.	integer

Responses

Code	Description	Links
200	<i>Content</i> application/json	No Links

Type	Name	Scopes
apiKey	jwtAuth	

POST /api/books/ Create a book

Creates a new book. Runs the 7-step cleaning pipeline automatically. Curator+ only.

Responses

Code	Description	Links
201	<i>Content</i> application/json	No Links

Type	Name	Scopes
apiKey	jwtAuth	

GET /api/books/hybrid-search/ Hybrid search with Reciprocal Rank Fusion

Blends PostgreSQL FTS relevance with Wilson Score popularity using Reciprocal Rank Fusion (Cormack et al., 2009).

RRF score = $\text{fts_weight} * (1 / (k + \text{fts_rank})) + \text{pop_weight} * (1 / (k + \text{pop_rank}))$ where $k=60$.

Returns rrf_score, fts_rank, and wilson_score per result.

Parameters

Type	Name	Description	Schema
query	q <i>required</i>	Search query	string

Type	Name	Description	Schema
query	popularity_weight <i>optional</i>	Popularity signal weight (default 0.3)	number (float)
query	page <i>optional</i>	Page number	integer
query	fts_weight <i>optional</i>	FTS signal weight (default 0.7)	number (float)
query	page_size <i>optional</i>	Results per page (max 100)	integer

Responses

Code	Description	Links
200	RRF-ranked results with rrf_score, fts_rank, wilson_score	No Links
400	Missing ?q= parameter or invalid weights	No Links

Type	Name	Scopes
apiKey	jwtAuth	

GET /api/books/recommendations/ Get personalised book recommendations

Returns personalised recommendations using a three-strategy waterfall:

1. **content_based** — genre/author profile from books rated ≥ 4
2. **collaborative** — Jaccard similarity with similar users
3. **popularity_wilson_score** — cold-start Wilson Score fallback

Response includes recommendation_strategy and profile_summary (top genres, top authors, total ratings).

Parameters

Type	Name	Description	Schema
query	limit <i>optional</i>	Number of results (default 20, max 50)	integer
query	strategy <i>optional</i>	Force strategy: content_based	collaborative

Responses

Code	Description	Links
200	Recommendations with strategy and profile summary	No Links

Code	Description	Links
401	Authentication required	No Links

Type	Name	Scopes
apiKey	jwtAuth	

GET /api/books/search/ Full-text search books

PostgreSQL full-text search using SearchVector + SearchRank with GIN index. Supports websearch syntax: "exact phrase", word1 word2 (AND), OR, -exclusion.

When ?q= is provided, returns results annotated with rank and headline (highlighted description snippet with <mark> tags).

Without ?q=, returns filtered list ordered by average_rating.

Parameters

Type	Name	Description	Schema
query	q <i>optional</i>	Search query (websearch syntax)	string
query	min_quality <i>optional</i>	Min quality score (0.0–1.0)	number (float)
query	published_before <i>optional</i>	Published year to	integer
query	author <i>optional</i>	Partial author match	string
query	ordering <i>optional</i>	Order by field (prefix - for desc)	string
query	min_rating <i>optional</i>	Min average rating (1–5)	number (float)
query	genre <i>optional</i>	Partial genre match	string
query	page <i>optional</i>	Page number	integer
query	max_quality <i>optional</i>	Max quality score (0.0–1.0)	number (float)
query	published_after <i>optional</i>	Published year from	integer
query	is_flagged <i>optional</i>	Filter flagged books	boolean

Type	Name	Description	Schema
query	page_size <i>optional</i>	Results per page (max 100)	integer

Responses

Code	Description	Links
200	Paginated search results with rank and headline	No Links
400	Invalid filter parameters	No Links

Type	Name	Scopes
apiKey	jwtAuth	

GET /api/books/{id}/ Retrieve a book

Returns full metadata for a single book by UUID.

Parameters

Type	Name	Description	Schema
path	id <i>required</i>	A UUID string identifying this book.	string (uuid)

Responses

Code	Description	Links
200	<i>Content</i> application/json	No Links

Type	Name	Scopes
apiKey	jwtAuth	

PUT /api/books/{id}/

Parameters

Type	Name	Description	Schema
path	id <i>required</i>	A UUID string identifying this book.	string (uuid)

Responses

Code	Description	Links
200	<i>Content</i> application/json	No Links

Type	Name	Scopes
apiKey	jwtAuth	

***DELETE* /api/books/{id}/ Delete a book**

Permanently deletes a book. Admin only.

Parameters

Type	Name	Description	Schema
path	id <i>required</i>	A UUID string identifying this book.	string (uuid)

Responses

Code	Description	Links
204	No response body	No Links

Type	Name	Scopes
apiKey	jwtAuth	

***PATCH* /api/books/{id}/ Update a book**

Partially updates a book record. Curator+ only.

Parameters

Type	Name	Description	Schema
path	id <i>required</i>	A UUID string identifying this book.	string (uuid)

Responses

Code	Description	Links
200	Content application/json	No Links

Type	Name	Scopes
apiKey	jwtAuth	

***GET* /api/books/{id}/my-rating/ Get my rating for a book**

Returns the authenticated user's rating for a specific book. 404 if not yet rated.

Parameters

Type	Name	Description	Schema
path	id <i>required</i>	A UUID string identifying this book.	string (uuid)

Responses

Code	Description	Links
200	The user's rating	No Links
404	User has not rated this book	No Links

Type	Name	Scopes
apiKey	jwtAuth	

***POST* /api/books/{id}/rate/ Rate a book**

Creates or updates the authenticated user's rating for a book (upsert). Rating 1–5. Rating ≥ 4 = upvote, ≤ 2 = downvote, 3 = neutral.

Triggers a Django signal that recomputes the book's average_rating, rating_count, upvote_count, and downvote_count.

Parameters

Type	Name	Description	Schema
path	id <i>required</i>	A UUID string identifying this book.	string (uuid)

Responses

Code	Description	Links
200	Rating updated	No Links
201	Rating created	No Links
400	Invalid rating value	No Links
401	Authentication required	No Links

Type	Name	Scopes
apiKey	jwtAuth	

***GET* /api/books/{id}/ratings/ List ratings for a book**

Returns all ratings submitted for a specific book.

Parameters

Type	Name	Description	Schema
path	id <i>required</i>	A UUID string identifying this book.	string (uuid)

Responses

Code	Description	Links
200	List of ratings	No Links

Type	Name	Scopes
apiKey	jwtAuth	

GET /api/imports/

POST /api/imports/ — upload CSV, triggers import, returns job GET /api/imports/ — list all import jobs (admin only) GET /api/imports/{id}/ — check status of a specific import job

Parameters

Type	Name	Description	Schema
query	search <i>optional</i>	A search term.	string
query	ordering <i>optional</i>	Which field to use when ordering the results.	string
query	page <i>optional</i>	A page number within the paginated result set.	integer

Responses

Code	Description	Links
200	<i>Content</i> application/json	No Links

Type	Name	Scopes
apiKey	jwtAuth	

POST /api/imports/

POST /api/imports/ — upload CSV, triggers import, returns job GET /api/imports/ — list all import jobs (admin only) GET /api/imports/{id}/ — check status of a specific import job

Responses

Code	Description	Links
201	Content application/json	No Links

Type	Name	Scopes
apiKey	jwtAuth	

GET /api/imports/{id}/

POST /api/imports/ — upload CSV, triggers import, returns job GET /api/imports/ — list all import jobs (admin only) GET /api/imports/{id}/ — check status of a specific import job

Parameters

Type	Name	Description	Schema
path	id <i>required</i>	A UUID string identifying this import job.	string (uuid)

Responses

Code	Description	Links
200	Content application/json	No Links

Type	Name	Scopes
apiKey	jwtAuth	

GET /api/ratings/

Dedicated endpoints for a user's own ratings.

GET /api/ratings/ — list all of my ratings GET /api/ratings/{id}/ — retrieve one of my ratings PATCH /api/ratings/{id}/ — update my rating DELETE /api/ratings/{id}/ — delete my rating

Parameters

Type	Name	Description	Schema
query	search <i>optional</i>	A search term.	string
query	ordering <i>optional</i>	Which field to use when ordering the results.	string
query	page <i>optional</i>	A page number within the paginated result set.	integer

Responses

Code	Description	Links
200	Content application/json	No Links

Type	Name	Scopes
apiKey	jwtAuth	

GET /api/ratings/{id}/

Dedicated endpoints for a user's own ratings.

GET /api/ratings/ — list all of my ratings GET /api/ratings/{id}/ — retrieve one of my ratings PATCH /api/ratings/{id}/ — update my rating DELETE /api/ratings/{id}/ — delete my rating

Parameters

Type	Name	Description	Schema
path	id <i>required</i>	A UUID string identifying this book rating.	string (uuid)

Responses

Code	Description	Links
200	Content application/json	No Links

Type	Name	Scopes
apiKey	jwtAuth	

DELETE /api/ratings/{id}/

Dedicated endpoints for a user's own ratings.

GET /api/ratings/ — list all of my ratings GET /api/ratings/{id}/ — retrieve one of my ratings PATCH /api/ratings/{id}/ — update my rating DELETE /api/ratings/{id}/ — delete my rating

Parameters

Type	Name	Description	Schema
path	id <i>required</i>	A UUID string identifying this book rating.	string (uuid)

Responses

Code	Description	Links
204	No response body	No Links

Type	Name	Scopes
apiKey	jwtAuth	

PATCH /api/ratings/{id}/

Dedicated endpoints for a user's own ratings.

GET /api/ratings/ — list all of my ratings GET /api/ratings/{id}/ — retrieve one of my ratings PATCH /api/ratings/{id}/ — update my rating DELETE /api/ratings/{id}/ — delete my rating

Parameters

Type	Name	Description	Schema
path	id <i>required</i>	A UUID string identifying this book rating.	string (uuid)

Responses

Code	Description	Links
200	Content application/json	No Links

Type	Name	Scopes
apiKey	jwtAuth	

GET /api/analytics/genre-trends/ Genre rating trends

Per-genre, per-month average rating trends. Uses TruncMonth to group BookRating records by calendar month. Admin only.

Parameters

Type	Name	Description	Schema
query	months <i>optional</i>	Lookback window in months (default 12, max 24)	integer

Responses

Code	Description	Links
200	List of (genre, month, avg_rating, rating_count) datapoints	No Links
401	Authentication required	No Links
403	Admin role required	No Links

Type	Name	Scopes
apiKey	jwtAuth	

GET /api/analytics/genre-quality/ Genre quality ranking

Genres ranked by average metadata quality score. Identifies which genres need curator attention. Admin only.

Parameters

Type	Name	Description	Schema
query	limit <i>optional</i>	Number of genres to return (default 10, max 20)	integer

Responses

Code	Description	Links
200	Genres with avg quality, book count, flagged pct	No Links

Type	Name	Scopes
apiKey	jwtAuth	

GET /api/analytics/summary/ Catalogue health summary

Full catalogue health snapshot: total books, flagged %, avg quality score, quality bands, genre distribution, rating statistics, top books by Wilson Score, import history. Admin only.

Responses

Code	Description	Links
200	Full catalogue health snapshot	No Links

Type	Name	Scopes
apiKey	jwtAuth	

Components

Schemas

Book

Properties

Name	Description	Schema
id <i>required</i>	<i>read only</i>	string (uuid)

Name	Description	Schema
title <i>required</i>	Maximum Length: 500	string
author <i>required</i>	Maximum Length: 300	string
isbn_13 <i>required</i>		string
description <i>optional</i>		string
genre <i>optional</i>	Maximum Length: 100	string
published_year <i>optional</i>	<i>nullable</i> Maximum: 2147483647 Minimum: -2147483648	integer
publisher <i>optional</i>	Maximum Length: 300	string
page_count <i>optional</i>	<i>nullable</i> Maximum: 2147483647 Minimum: -2147483648	integer
language <i>optional</i>	Maximum Length: 10	string
cover_url <i>optional</i>	Maximum Length: 200	string (uri)
normalized_title <i>required</i>	<i>read only</i>	string
normalized_author <i>required</i>	<i>read only</i>	string
quality_score <i>required</i>	<i>read only</i>	number (double)
is_flagged <i>required</i>	<i>read only</i>	boolean
genre_confidence <i>required</i>	<i>read only</i>	number (double)
average_rating <i>required</i>	<i>read only</i>	number (double)
rating_count <i>required</i>	<i>read only</i>	integer
upvote_count <i>required</i>	<i>read only</i>	integer
downvote_count <i>required</i>	<i>read only</i>	integer

Name	Description	Schema
vote_distribution <i>required</i>	<i>read only</i>	object
created_by_username <i>required</i>	<i>read only</i>	string
created_at <i>required</i>	<i>read only</i>	string (date-time)
updated_at <i>required</i>	<i>read only</i>	string (date-time)

BookList

Lighter serializer for list views — fewer fields for performance.

Properties

Name	Description	Schema
id <i>required</i>	<i>read only</i>	string (uuid)
title <i>required</i>	Maximum Length: 500	string
author <i>required</i>	Maximum Length: 300	string
genre <i>optional</i>	Maximum Length: 100	string
isbn_13 <i>optional</i>	<i>nullable</i> Maximum Length: 13	string
published_year <i>optional</i>	<i>nullable</i> Maximum: 2147483647 Minimum: -2147483648	integer
quality_score <i>optional</i>	Maximum: 1.0 Minimum: 0.0	number (double)
is_flagged <i>optional</i>		boolean
average_rating <i>optional</i>		number (double)
rating_count <i>optional</i>	Maximum: 2147483647 Minimum: -2147483648	integer
cover_url <i>optional</i>	Maximum Length: 200	string (uri)
normalized_title <i>optional</i>	Maximum Length: 500	string

Name	Description	Schema
normalized_author <i>optional</i>	Maximum Length: 300	string

BookRating

Properties

Name	Description	Schema
id <i>required</i>	<i>read only</i>	string (uuid)
book <i>required</i>	<i>read only</i>	string (uuid)
book_title <i>required</i>	<i>read only</i>	string
rating <i>required</i>	Maximum: 5 Minimum: 1	integer
review <i>optional</i>		string
vote_type <i>required</i>	<i>read only</i>	string
username <i>required</i>	<i>read only</i>	string
created_at <i>required</i>	<i>read only</i>	string (date-time)
updated_at <i>required</i>	<i>read only</i>	string (date-time)

BookRequest

Properties

Name	Description	Schema
title <i>required</i>	Maximum Length: 500 Minimum Length: 1	string
author <i>required</i>	Maximum Length: 300 Minimum Length: 1	string
isbn_13 <i>required</i>	Minimum Length: 1	string
description <i>optional</i>		string
genre <i>optional</i>	Maximum Length: 100	string

Name	Description	Schema
published_year <i>optional</i>	<i>nullable</i> Maximum: 2147483647 Minimum: -2147483648	integer
publisher <i>optional</i>	Maximum Length: 300	string
page_count <i>optional</i>	<i>nullable</i> Maximum: 2147483647 Minimum: -2147483648	integer
language <i>optional</i>	Maximum Length: 10 Minimum Length: 1	string
cover_url <i>optional</i>	Maximum Length: 200	string (uri)

CustomTokenObtainPairRequest

Properties

Name	Description	Schema
username <i>required</i>	<i>write only</i> Minimum Length: 1	string
password <i>required</i>	<i>write only</i> Minimum Length: 1	string

ImportJob

Properties

Name	Description	Schema
id <i>required</i>	<i>read only</i>	string (uuid)
status <i>required</i>	<i>read only</i>	
file_name <i>required</i>	<i>read only</i>	string
total_rows <i>required</i>	<i>read only</i>	integer
cleaned_count <i>required</i>	<i>read only</i>	integer
duplicate_count <i>required</i>	<i>read only</i>	integer
failed_count <i>required</i>	<i>read only</i>	integer
error_log <i>required</i>	<i>read only</i>	

Name	Description	Schema
progress_percent <i>required</i>	<i>read only</i>	number (float)
created_by_username <i>required</i>	<i>read only</i>	string
created_at <i>required</i>	<i>read only</i>	string (date-time)
completed_at <i>required</i>	<i>read only</i> <i>nullable</i>	string (date-time)

PaginatedBookListList

Properties

Name	Description	Schema
count <i>optional</i>		integer
total_pages <i>optional</i>		integer
current_page <i>optional</i>		integer
next <i>optional</i>	<i>nullable</i>	string
previous <i>optional</i>	<i>nullable</i>	string
results <i>optional</i>		< BookList > array

PaginatedBookRatingList

Properties

Name	Description	Schema
count <i>required</i>		integer
next <i>optional</i>	<i>nullable</i>	string (uri)
previous <i>optional</i>	<i>nullable</i>	string (uri)
results <i>required</i>		< BookRating > array

PaginatedImportJobList

Properties

Name	Description	Schema
count <i>required</i>		integer
next <i>optional</i>	<i>nullable</i>	string (uri)
previous <i>optional</i>	<i>nullable</i>	string (uri)
results <i>required</i>		< ImportJob > array

PatchedBookRatingRequest

Properties

Name	Description	Schema
rating <i>optional</i>	Maximum: 5 Minimum: 1	integer
review <i>optional</i>		string

PatchedBookRequest

Properties

Name	Description	Schema
title <i>optional</i>	Maximum Length: 500 Minimum Length: 1	string
author <i>optional</i>	Maximum Length: 300 Minimum Length: 1	string
isbn_13 <i>optional</i>	Minimum Length: 1	string
description <i>optional</i>		string
genre <i>optional</i>	Maximum Length: 100	string
published_year <i>optional</i>	<i>nullable</i> Maximum: 2147483647 Minimum: -2147483648	integer
publisher <i>optional</i>	Maximum Length: 300	string
page_count <i>optional</i>	<i>nullable</i> Maximum: 2147483647 Minimum: -2147483648	integer
language <i>optional</i>	Maximum Length: 10 Minimum Length: 1	string

Name	Description	Schema
cover_url <i>optional</i>	Maximum Length: 200	string (uri)

PatchedUserProfileRequest

Properties

Name	Description	Schema
username <i>optional</i>	Required. 150 characters or fewer. Letters, digits and @/./+/_/ only. Maximum Length: 150 Minimum Length: 1	string
email <i>optional</i>	Maximum Length: 254 Title: Email address	string (email)
bio <i>optional</i>		string

RegisterRequest

Properties

Name	Description	Schema
username <i>required</i>	Required. 150 characters or fewer. Letters, digits and @/./+/_/ only. Maximum Length: 150 Minimum Length: 1	string
email <i>required</i>	Maximum Length: 254 Title: Email address	string (email)
password <i>required</i>	<i>write only</i> Minimum Length: 1	string
password2 <i>required</i>	<i>write only</i> Minimum Length: 1	string
role <i>optional</i>		RoleEnum

RoleEnum

- **reader** - Reader
- **curator** - Curator
- **admin** - Admin

StatusEnum

- **pending** - Pending

- **processing** - Processing
- **completed** - Completed
- **failed** - Failed

TokenRefresh

Properties

Name	Description	Schema
<i>access required</i>	<i>read only</i>	string

TokenRefreshRequest

Properties

Name	Description	Schema
<i>refresh required</i>	<i>write only</i> Minimum Length: 1	string